

ActInf GuestStream 082.7 ~ Robert Worden "Language and Human Nature"

GuestStream | Robert Worden | 1:21:31

Hello, welcome everyone. It is...

You want me to start, right? Sorry. Oh, can I go now?

You can.

Yeah, right.

Alright.

Okay.

Hi, this is Roland Worden here, presenting a second live stream about this unified theory of language, which is described in an archive free print, which you can see at that address.

So this is the second live stream. In the first...

And this is a slide I showed the first time around about the various bits of language we just studied.

Phenology, syntax, semantics, pragmatics, learning and evolution.

And the two bold ones, pragmatics, how we converse, and evolution, how language evolved.

Those are the bits we're going to focus on this time.

And I should say that these two columns, speaker and listener, are put in there just to remind us that you have to...

For all of these problems, you have to pay attention to both how the speaker goes about it and how the listener goes about it.

And I'll try and do that while we're going through this.

So, just a few key points from part one of the live stream.

And these are the points most relevant to this second part.

I said the language is hierarchical based in pattern matching, which is very much in line with the FEP.

And so you could have an FEP implementation language, if you'd like to, although that hasn't been done, to my knowledge.

The approach I'd be taking here is CHOP from cognitive linguistics, or what's also called construction grammars.

It's not a Chomskin approach.

And in cognitive linguistics, there are thousands of patterns active in language.

And in fact, there's at least one pattern for every word in a language.

So that makes thousands of patterns.

And cognitive linguistics has produced working models of analogy, syntax, semantics, and language learning. It hasn't done so much about pragmatics.

Generally, the study of pragmatics has lagged long behind the study of the other subjects like semantics and syntax.

So in cognitive linguistics, every word is represented by what they call a construction.

That is a feature structure, which is a directed acyclic graph structure.

And I showed some examples of these.

And the key operation in language in these construction-based grammars is unification.

And I showed that unification is a form of Bayesian pattern matching.

So that's the key points from the first live stream.

What this live stream is going to be about is mainly about pragmatics and evolution.

I should start by saying language long ago started in conversation.

And we had conversations before we had syntax and semantics.

Language evolved by sexual selection for the display of intelligence.

That's how language got started.

And because of that, language is a biased thing.

Language is not, as some people imagine, a kind of neutral format for talking about reality.

It's actually biased to be all about social status.

And I'll try to illustrate that.

Now, coming back to pragmatic conversation, conversation requires mind reading.

That's the essential requirement to carry on a conversation.

And then I shall present a model of pragmatic, model of conversations,

which is built on Bayesian pattern matching, just like the syntax and semantics we talked about in the last live stream.

And so the basis of this is that we learn mind reading patterns.

And in those mind reading patterns, we often, when we're thinking, we often have what I call a shadow audience,

which we're trying to work out what the shadow audience will think about what we say and what we do.

So that shadow audience is a kind of something we imagine, which assesses our own status.

And the shadow audience is the basis of the human self.

And so the shadow audience is very important because it's the basis of all human self is the basis of all societies, how we collaborate.

But also it has a dark side. And that is, I shall describe the main affliction of mankind.

And finally, I should talk a little about mindfulness, which is a release from the human self and the shadow audience.

So that's the main topic that I'm going to discuss in this live stream.

Just get straight into it. Language started in conversation.

Imagine about a million years ago, before we had semantics or syntax, we started to converse with each other.

The semantics part of it was by mimicking, by gesture and by pointing.

You point a thing, draw attention to them and so on.

And for all conversation between two people, there has to be a common ground of shared understanding between those two people.

And so a conversation starts with two people both understanding something about their current situation.

And each turn in the conversation, one person, then another person, each turn adds some facts of the

common ground.

And some of these facts are about the mental states of the speakers.

And I shall illustrate this in a minute.

So because of that, conversation requires mind reading, requires each person to infer things about the other person's mental state.

And that we do by Bayesian pattern matching.

And so a very important part of pragmatics is the study of speech acts.

Speech acts are ways in which one party in a conversation adds facts about his own mental state to the common ground.

And just to give an example of this, a very early conversation, say a million years ago, between two people, A and B.

And they have a certain common ground, the red points before the conversation starts.

The common ground is that time is the afternoon.

A is a man who's just been hunting.

B is a son.

There's a fire that they can both see.

And there is some meat because A brought it back from hunting.

And they both have some general knowledge about the world.

They both know that to eat some meat, you have to cook it.

So that stuff above the dashed line is the common ground at the start.

And the conversation between them, I show it is illustrated by these points here, which each add facts to the common ground.

So A says eat and he points to something.

And so he's offering that B might eat something.

That's an offer speech act.

B says yes.

And so B accepts the offer of A.

And then B says you cook.

And B wants to know who's going to do the cooking.

So B wants B a state in which B knows who is going to do the cooking.

So you can see here, we're building up facts about both parties' mental states.

A says yes.

So that adds the fact that A is going to do the cooking.

So those are the facts which are explicitly added to the common ground.

There are other facts which can be deduced from that.

Because they're in the common ground, B knows that A is going to do the cooking.

A knows that B knows that A is going to do the cooking.

So there are loads of mental state facts in the common ground.

And the conversation proceeds by speech acts, which have different kinds, like offers and questions and answers and so on,

which add facts to the common ground.

So that's the basic model of a conversation.

How conversations started a long time ago, before we had elaborate syntax or semantics, before most of language.

So in this picture, semantics came about by making speech acts more fine-grained.

There are a few broad types of speech act, which are declarations where you say something and offer a request or a command or a question and so on.

And each speech act adds some facts about your own mental state to the common ground.

Now semantics is how we develop language to allow speech acts to be more fine-grained.

They can be about certain agents or acts or patients.

They can be at different times or different attitudes to them and so on.

So that's how semantics came about.

And syntax is just the structure of language we need to convey semantics.

So in this picture, syntax and semantics came out of pragmatic.

Pragmatic came first.

Now that, interestingly, is quite the reverse of the order in which language has been traditionally studied.

There's been a long-term study of syntax and semantics starting way back when.

And pragmatics has always been a kind of poor relation that came later.

But in this picture, pragmatics actually came first.

So what is pragmatics?

It's studied under a number of headings and these on the left-hand side are the traditional headings under which we've studied pragmatics.

And here are some examples of it.

Here are some speech acts, a greeting, hello, there's a request, can you pass the salt?

And the same sort of structure but a very different thing is a question, can you pole vault?

In other words, this wants the listener to do something.

And this wants the listener to just provide some information.

And here's an answer, no.

Reference resolution, sometimes called deixis.

In other words, we're always using noun phrases, nouns, referring to things.

We have to work out what thing precisely is being referred to.

So you use the word him.

Somebody's got to work out what him refers to.

Or you talk about the blue one near the back.

Somebody has to work out what that is.

Now, there are a couple of topics which are commonly studied, which are quite closely related.

Presupposition and implicature.

Both of them, you're actually beyond the obvious thing that you're saying, there's something else added to the conversation, to the common ground.

Presupposition is when you say something, it presupposes something else.

You say, my cat's not hungry.

That presupposes you've got a cat.

When you say he regrets going to Spain, that kind of presupposes he actually did go to Spain.

Implicature is an implication of what you say.

You say, shall we ask him?

Oh, he's got an essay to write.

And the listener has to work out.

That means he can't come along.

Or if you say, I caught some of the geese.

You're somehow implying that you didn't catch all of the geese.

So those are presuppositions and implicature.

Then there are other topics like irony, where you say, yeah, sure.

Or that's really brilliant.

Or hyperbole, exaggerate, you're a genius.

Or metaphor, where you say, she's a brick.

In other words, she's a very reliable person.

Or you say, he's a chocolate teapot.

You're saying the opposite.

He's an unreliable person.

So those are some of the subjects in pragmatics.

And the model of pragmatics has to give some account of all of them.

And you can see if you look at them, look at each example.

There is the common ground features in how you work out what's going on and mind reading features as well.

What I've developed recently is a Bayesian model of pragmatics.

And I'll just try and describe in outline how it works.

So both people in their heads have a version of the common ground.

They start with a shared understanding of what's going on.

And each fact in the common ground has a think-all-out, which I call prominent.

A very prominent fact is the most recent thing added to the common ground, as well as a Bayesian probability.

And if some facts you can infer from other facts, generally the inferred facts are less prominent than the things they're inferred from.

So in this model, there are pragmatic constructions, which are just like the word constructions of syntax and semantics that I talked about last time.

So facts in the common ground are created by unifying various things together.

The sound of the utterances that you hear or are going to speak, pragmatic constructions and prior facts in the common ground.

These unifications are always done in an order, prominence first until you get a good enough result.

And unification, as I said last time, is Bayesian pattern matching.

So pragmatic constructions are just like the other constructions of language like word constructions.

They have a phonetic pole, which includes the sound part and the semantic pole and may include other things as well.

And a semantic pole, which includes only the meaning.

So what happens in a conversation is that the speaker and the listener unifies the same pragmatic construction.

The speakers unify in one direction from the semantic pole to the phonetic pole and listeners go in the opposite direction.

They hear sounds that they go from the phonetic pole to the semantic pole.

So this is a version of both how the speakers work out what to say and how the listeners work out what they mean.

And as a result of this in the model, you can show how both versions of the common ground, the speaker's version and the listener's version, stay in step.

And so this is a model in which semantics comes out of the pragmatics.

Now, this model of pragmatics is different from the way pragmatics has been studied in the past.

Previous work on pragmatics has been in two different schools.

This so-called Neo-Grythian school, which is named after the philosopher Paul Grice, who really started the study of pragmatics in about 1960s.

And the names associated with the Neo-Grythian school are Horne and Stephen Levinson and various others.

And the other school of pragmatics is called relevance theory, which is started by Dan Sperber and I and Deirdre Wilson.

Now, both of those schools take the traditional view that semantics comes first and the mind, the listener or the speaker does a bunch of semantics first and then chucks the result over the wall, if you like, to a pragmatic inference part of the brain.

And both schools place all their emphasis on the work has to be done by the listener to understand what he hears.

And they both practically ignore the problem of the work done by the speaker to work out what to say next to be understood.

So they're very not sighted.

The other thing about those models is they describe the pragmatic inference in words, in verbal descriptions, and they don't have really working models.

So the advantages of this approach over previous approaches is there's no fixed boundary between pragmatics and semantics.

Two are intermingled all the way.

This is a joint model that solves both sides of the problem, the speaker's problem and the listener problem.

And it's a continuous extension of the Bayesian model that I described in the last live stream.

And it's an extension of cognitive linguistics.

And the final advantage is it works.

And it works fast because one thing in our conversations is that we can respond in conversation very fast.

And there's a puzzle about how we do all the inferences we required so rapidly.

In this model, the inferences are done pre-consciously by Bayesian pattern matching just by like so many other things in the mind.

Right. So there's the model of pragmatics.

I shall now skip to a different subject to come back to pragmatic later.

How did language evolve?

I discussed this to a certain extent in the last live stream.

And in the first live stream I said language uses hierarchical Bayesian pattern matching like so many things in the brain.

And this talk addresses part two.

How did language evolve particularly in mankind and in no other species?

So that is the next few slides.

I shall first talk about natural selection and sexual selection.

As we well know, to pass on your genes, you need to do two things.

You need to survive until you're an adult and you can reproduce,

and you need to find a mate in order to reproduce.

So these are really different kinds of pressures.

They lead respectively to natural selection,

how you survive to adulthood, and sexual selection,

how you find a mate.

Now sexual selection is a very common force in nature.

It leads to the kind of displays that we see in flowers and birds,

plumage and all sorts of things.

It leads to the diversity of species on the planet,

and it leads to sexual dimorphs and the differences between males and females.

So if I say language involved evolution involved sexual selection,

that's really not a remarkable thing to say.

It would be much more remarkable if sexual selection were not involved

in the evolution of language.

So the hypothesis is, which I published in Frontiers Journal,

that language evolved...

I'm sorry, I'll talk a little more about how sexual selection works first.

Natural selection is a gradual adaptation to the habitat.

So it's a hill climbing process, an optimization process.

Sexual selection is very different.

The maths of sexual selection worked out about 1980 by Landay and by John Maynard Smith.

And the maths is a positive feedback process, positive feedback between genes in the male and genes in the female.

And the results of sexual...

of this mathematical model, which agree with the data, are sexual selection leads to novel and vivid traits, just like the peacock's tail shown here.

Fast visible changes of phenotype.

Again, the peacock's tail is very visible because it has to be visible to get a male find a mate.

And the results of sexual selection are always unique to one species.

The peacock's completely different from any other species of birds.

The results of sexual selection are always exaggerated, strongly exaggerated beyond what is most fit.

So the peacock's tail obviously is a big handicap for the male peacock, but it needs that to get a mate.

So the final point is they are handicapped.

So these are the results of sexual selection.

Novel, fast changes, unique to a species, exaggerated and handicaps.

So here's the hypothesis, which I published about a couple of years ago in *Frontiers*, that language evolved by sexual selection, particularly to display intelligence.

In order to get a mate, we have to show off our intelligence.

And that, as I say, is a very unremarkable hypothesis, but it does answer the big questions of language evolution.

So here are some of the questions and the answers.

Why do we have such ludicrously large brains?

The answer is there is sexual selection to display intelligence.

And to display intelligence, you always have to have it.

Why language? Why do we display intelligence by language?

Because words is a very quick and efficient way to show off our intelligence.

Why is language so amazingly fast and expressive?

In fact, it's much faster than we would need in a natural habitat for any natural purpose, like arranging hunts or arranging gathering or whatever, planning a camp.

Language can say things in a few seconds, whereas for practical purposes, you might be able to take a few minutes.

And language is so fast because sexual selection causes exaggerated traits.

The speed of language has been exaggerated beyond the biological need.

A big question about language evolution is why do closely related species, particularly chimps, not have language like ours?

They're so different.

And the answer is simply sexual selection is unique to a species.

It's not a habitat selection pressure.

It's a species internal selection pressure.

And the final question, why do we have language when the benefits of language don't match the huge costs of our extra brain, our large brain, 20% extra food requirement?

And the answer is sexual selection causes handicaps.

So all in all, it seems very likely that language did evolve in this process by sexual selection and it is a tool for doing practical things.

We do practical things with it.

Primarily, it's a tool for showing off our intelligence.

Now, intelligence is related to social status.

We show off our intelligence using language as a way to get a mate.

But getting a mate also requires high status in the social group.

So when we're using language, we're always very concerned about our own social status.

And what this means is that when we're doing conversation using these pragmatic mind reading patterns, we use those patterns to infer not only what the other person knows about the world in general, but what they think about us.

In other words, in a conversation, we're always asking ourselves questions like,

Will he think I'm impolite?

Will she think I'm being uncaring?

And these are all to do with our social status.

So this is a very powerful pre-conscious filter of what we say.

In other words, we often ask these questions.

We say, no, I better not say that.

But it's done very quickly because there's a famous result from Levent and Torreira that people take turns in conversations within a fifth of a second of the other person's stopping.

They work these things out very rapidly.

The way they work them out rapidly is by pre-conscious Bayesian pattern matching with these so-called status patterns.

And we have hundreds of those status patterns.

If I say this, they will think I'm good or bad, Torreira maybe.

So intelligence, showing off intelligence in language is very closely related to maintaining our social status.

Language is about social status.

We can see that not from the evolution of language, but from the present day properties of what language looks like now.

We can take various aspects of language and show how they are relevant to status.

The most basic aspect of language is the structure of a sentence.

Agent, act, patient or agent, act.

And this kind of implies that the agent has chosen to do something.

The agent is exercised in free will and will be judged by the results of those actions.

The contrast with animals is animals, of course, have no free will.

They just do what the moment requires.

So animals don't make choices.

Language obviously has future tense and past tense.

And why was that introduced?

Because people's status can be judged, not only by what they're doing now, but what they did in the past and what they will do in the future.

And again, there's a big contrast with the animals that we evolved from.

Animals just live in the present.

I think their minds have little or no use for future episodes or past episodes.

So animals would not be able to use the future or past tense.

Possession language has words like mine, your and so on.

They're introduced because possession is control and control has status implications.

If I've got a fast car, if it's mine, then that increases my status.

Some people think.

Animals, of course, don't have possessions.

There are modal verbs in language, verbs like can and should and would and so on.

Can implies that you've chosen that you can choose to do things or not to do things.

Should implies society thinks you ought to do it.

It's a good action.

So both those, all those modal verbs are laden with status and implication.

Whereas animals don't choose what to do and they don't modal verbs are not relevant to them.

Can't say an animal can do this.

Animal either does this or it doesn't do that.

We have mental state verbs, knows and wants and intends.

Again, states of mind are very relevant to status.

If somebody has bad intentions, that's a low status thing to be.

Language has comparisons, bigger, better and so on.

Comparison is relevant to social ranking.

In other words, bigger and so on.

It's always putting people in order of social ranking.

So language is absolutely soaked with features which are all to do with social status.

It's not neutral.

And with every utterance, we pre-consciously infer what it implies about our own status in the eyes of other people.

Using these learned status patterns.

And that brings me onto an important topic called the shadow audience.

When we talk to somebody, we're pre-consciously inferring what that person will think about us.

We're matching these status patterns to filter what we say.

But when we're on our own privately thinking, that evolved as a rehearsal of what we're going to say.

That's thinking is rehearsing what we might say to somebody.

And we'd rehearse in order to make it more impressive.

But when you're privately thinking, there is no real audience there.

So what we do is we invent shadow audiences.

People we know who we imagine how they will judge what we say.

They could be parents or friends or just people in general.

We use these status patterns that we learned in conversation to infer what the shadow audience is going to think of us.

And this is a very important part of our thought process.

It's going on the whole time and it makes us feel emotional reaction.

In other words, if I think this is going to make me look bad, I start to feel bad.

And so the shadow audience is actually what creates the human idea of myself.

That's the agent self.

That's the myself who's going to act and who's going to be judged by other people on the basis of what I say and do.

So the shadow audience is the thing that really creates the human sense of self, which is much more elaborate and complex than any animal sense of itself.

So with that in mind, we can consider how thoughts work in fractions of a second.

And there you are thinking, hearing the words in your mind, and you may be hearing about two or two, three or four words every second.

And so your brain is doing some very fast pre-conscious pattern matching and matching two to four word patterns per second just to make up the sentence.

And it's matching those status patterns as well to think what will people think of me.

And the status patterns make you feel emotions.

And what happens when you feel an emotion is that bodily feeling may match some further status patterns into cascades of status patterns.

And this is all happening faster than conscious thought.

So that's why we feel such irrational and unpredictable emotions sometimes.

And it comes to the quote from Blaise Pascal,

Le peur à ses raisons que la raison ne connaît pas.

The heart has its own reason which reason doesn't understand because it's happening faster than conscious reasoning.

So this leads to what I call the dark side of language.

These learned status patterns are part of language and they enable us to cooperate with each other as members of society.

So this is what has underpinned the whole of human progress, why we conquer the planet.

But these status patterns do have a dark side.

We are constantly feeling anxious that we should be doing better.

And we have this idea of myself as other people think about me, but it's a second-hand idea.

It's not what I think of me.

It's what I think other people think.

And there's a flawed idea because the patterns we use aren't perfect.

They're very inaccurate sometimes.

So we are constantly looking at our own life history to polish it and refine it to preserve our self-esteem.

We feel these emotions in chaotic ways.

And to keep our shadow audiences happy, we agree with what we think they're going to think.

In other words, we don't think for ourselves.

We think, what would the group think about this?

That's group think.

And the second part of that is to enhance our own status.

We have out-groups.

We have other people who, in other groups, who would diminish them and we diminish their status.

So this is what leads to tribalism and intolerance.

So the dark side of language is the root of human affliction and evil.

It's produced some great things in society and civilization, but some very bad things, which we're all aware of.

So next I come onto the topic of meditation and mindfulness.

Recapping what I've said before.

The human agent self, the self which acts and which is judged by other people, is learned in language in these status patterns.

Now, I believe the mindful states are states in which that agent self is diminished or absent.

This agrees very well with the phenomenal reports of what it's like to be in mindful state.

And there's a very good collection of these phenomenal reports in a book by Metzinger called *The Elephant and the Blind*.

And we can go through these reports of what mindful states are like in Metzinger.

And we can interpret them in terms of the hypothesis that it's all about language status patterns.

So meditative states are relaxed and peaceful.

And the reason is that the anxiety in the cascades of thought, the status patterns create, are reduced in meditative states.

There's a silence of non-reaction about it.

And when there's less language going on, there are few active status patterns.

And even when thoughts arise, they can be just observed as thoughts without reactions.

There's a sense of non-identification, no self, no physical boundaries and the connectedness with other things.

And the interpretation here is that the agent self is created by the status patterns and creates the boundary between the self and the rest of the world.

And so that boundary gets dissolved.

There is a lot of compassion in meditative states.

And the interpretation is that when the agent self is not protecting its own status, we are more open to seeing what life is really like for other people,

than seeing the world from their point of view.

In meditation, there is no time.

In deep meditation, there's no time and there's no life stories.

Chronological time has been created by language, driven by status concerns.

And we build these life stories to protect our own status.

When we're not concerned with our own status, we're not concerned with life stories.

There is a sense of clarity in what the meditations call suchness in meditation.

And what this is, is when we don't classify things according to whether they're good or bad for us, we can look at them more closely and see what they really are.

There's a sense in meditation sometimes of nothing left to do.

What that means simply is that the agent self is not present, created by the status patterns.

There's no need for action.

As meditation is associated with non-duality, the duality between self and other is brought about by language.

And when language is less active, there's less duality.

So that is the correlation between reports of what meditation is like and the theory that it's really a lack of language,

and the lack of the status patterns that makes it like that.

So we can approach this from another angle.

Some of the traditional things we know about Buddhism, many aspects of Buddhism we can interpret from this point of view.

The first noble truth of Buddhism is Dukkha, suffering.

But what this is, is continual discontent from the action of status patterns saying, we should be doing better.

When they're absent, Dukkha is absent.

The Buddhist cycle of birth and rebirth is interesting.

It doesn't mean we reincarnate physically, literally reborn.

What it means is that there's an agent self, which language is created, and it recreates itself in cascades or cycles.

The shadow audience creates the self and the self creates the shadow audience.

So that cycle gets abolished.

The nirvana, or non-duality, is when that cycle is no longer there,
allowing the non-dual experiencing self to be there.

Buddhism often uses the word conditioning.

That simply means the learning of status patterns that we learnt in language.

The one flavor of Zen Buddhism is probably the most strident rejection of language.

Zen Buddhism says you can't describe any of this in language at all.

So that interpretation of mindfulness accords with long known teachings of Buddhism.

Zen Buddhism says you can't learn a word.

Now, here is a hypothesis about how we learn mindfulness.

We know from language that we never unlearn a word.

Once you've learnt a word, we've got it for life, really.

So how can, if we've got these status patterns and they're the problem,
how can we unlearn them?

We can't unlearn the status pattern.

So how can we reduce their influence on us?

Well, there is a way because basic pattern matching is always competitive.

It's always several kinds of patterns are tried out

and only the one that wins the competition goes on to have a conscious effect.

The patterns that lose the competition don't have conscious effects.

So here's the hypothesis about how we learn mindfulness.

the status patterns, the status patterns that drive our emotions are triggered by feelings on our bodies.

But when meditating, what we do is we observe those bodily feelings as closely as we can.

And while we're doing that, as a side effect, we learn some new patterns,
which are called mindful patterns.

And these new patterns say when you see, when you feel certain bodily feelings,

like something you're breathing or something,

a stomach or whatever it may be,

simply pay closer attention to them, find out what they're like.

So this is a different pattern from the status pattern.

And so what happens is that mindful patterns compete against the status patterns
and win the competition.

So the status patterns are less influential when you pay attention to the mindful patterns.

So mindful states happen when status patterns are diminished or absent.

So there is a hypothesis about how meditation is learned.

Now, finally, last slide really, is a switch of topic,

something that I've postponed for some time, but an important aspect of language in all ways.

Every language consists of a very large number of constructions,
word constructions and status instructions and so on.

Each one of them is like a species and the whole language is like a habitat and ecology.

And one species reproduces by being used and learned and passed on.

So this is what happens.

People use a certain word, children hear it and they learn it, they pass it on.

And in that evolution, so that process evolves constructions.

Constructions compete with each other, compete against other constructions
and there are very strong selection pressures on the construction and the language.

They should have a useful meaning,
that people actually need to speak them, need to say them from time to time.

They must be brief enough.

If somebody can say things quickly, they'll do it the other way.

They must be unambiguous enough and they must be learnable.

So there are selection pressures on constructions, on words and status patterns,
which cause them to evolve.

They lead to very fast language change.

That's why language changes within a few human generations.

And what that means is that much of the structure we see in language is the result of that fast evolution.

And it's not the evolution of the human brain that made it that way.

For instance, there are language universals about the structure of most languages.

And they are telling us how languages have evolved.

And what that means is that language is actually telling us less about the human brain than we thought it did.

Okay, so that's the end of the second live stream of the prepared material.

There are various topics you might like to discuss.

How mind reading patterns operate in conversation.

Whether language did evolve by sexual selection.

How language is about status.

Status patterns use a shadow audience.

The human sense of self based on the shadow audience of being a rather second hand idea of ourselves.

Language and human affliction.

Mindfulness and evolution of language balance.

So, over for any questions.

Thank you, Robert.

There's a few things I wrote down and then I think this is a great menu.

And I'll also look in the live chat if anyone writes a question.

Okay, so.

Lot to think slash say in that self-aware language nexus.

I was wondering what could or would address that shadow side of language.

Is that a feature of some of the constraints of a given language?

Like reification of processes or objects or affixing adjectives in some way that somehow disrespects the ontological aspects of properties of systems?

Or is it like any symbolic machine or any symbolic approximation is going to have a shadow?

Or what is the alternative?

Or is it that shadow is a necessary follow?

It's a consequence of proposing symbolic representation of the ecological scenario.

Well, I think it came very early on.

In other words, we had shadow audiences before we had much semantics before we had much syntax before we had much symbolic systems.

You go back to the very early conversations.

Let me just go right back.

Conversation before people had many words, they just used pointing and took turns in conversation.

I think even then, people had real conversations in which they imagined what the other person thought of them.

So the real audience was there and people imagined conversations, rehearsed conversations.

So they had shadow audiences.

So I think the issues of shadow audience and status go way back beyond symbolic systems and beyond syntax and semantics.

So does it exacerbate or ameliorate the fundamental non-linguistic game theory of conformity and group living?

I think it's the root of it.

I think it's the origins, where it all came from.

But I think we wouldn't be-

Group living was prior to vocalization, though not necessarily prior to pheromonal or stigmatic communication.

Yeah, that's a very interesting question.

Whether that, I mean, whether that actually preceded language evolution.

In other words, in a chimpanzee group, they are conscious of their own status, probably.

They know some other chimps have higher status than them.

But I think this all became exaggerated in humans because of the sexual selection for showing off intelligence.

It became so much more complicated because we added intelligence to the mix as the thing that was sexy and attractive.

And that made it all so much more complicated.

Hmm.

Okay.

Where do you think this relates to a priori languages or to our sort of current cognitive linguistically informed

use of language?

How do you see this propositional explication of the role of language in cognition and evolution reentering into human language use?

As opposed to language being a useful tool, but not necessarily itself theorized about?

I'm not sure I get the question.

Like, do you think this will lead us to constructed languages or to elaboration or simplification of languages or some other way to think about how media and different formats of communication are used or norms around required or forbidden methods of communication based upon a grounded understanding of what the consequences of those violations of those norms would be?

Well, I think I don't see much future for synthetic languages used by people.

You know, I mean, people develop language, languages evolve as in the previous slide to be useful people.

And so I don't see that synthetic languages will do anything except evolve to be like ordinary languages.

Am I answering the question?

Probably not.

I think, I mean, that's kind of a funny conclusion.

But yes, the goal of a constructed language would be to become an ordinary one.

And how would you relate it to language of thought?

And if we have the cessation, whether we take this in a more meditative direction or not, what is in the absence of verbal or linguistic mental chatter?

What is the structure of thought that occurs without language?

Great question.

Well, perhaps you should look at medicine in his book to what he thinks goes on when there's not language happening.

But I think we always have a physical imagination.

We always can physically imagine situations and run simulation and work out what happens next.

So that's a component of thought which is going on even when language is going on.

I think everyday thought is a mixture of physically imagining things and putting words around it.

And the physically imagining things still goes on when you don't have words.

Physically meaning the situational and spatial awarenesses?

Spatial awareness, yeah.

How would you think the analog wave or spatial representations interface mechanistically or just cognitively with that naming process?

Is there a supervisor of that spatial wave that names or is the wave become translated into some describable process?

Well, if you'd say, if you believe as I do, the basis of consciousness, spatial consciousness is this wave in the thalamus, then that happens in all higher animals.

And any higher animal can take its spatial experience and can find things in the space around it and classify them.

You know, you can classify here's an insect, here's a flower, whatever.

So that classification into categories that are useful for the animal has evolved into classification into things that we name as human speakers.

So it goes back to animals.

But the animal brain has always been able to link the wave with classifications or things that are useful to know about.

Okay, I'll read some comments in the chat and we can look at these and some of these other topics.

I really enjoy the, Robert.

You're bringing a huge amount of inquiry and research to bear.

And it just lightly touches on a lot of key areas that brings a conciliance to a lot of the cognitive and linguistic quest, which is just, it's the water we swim in, metaphorically speaking.

Okay, Active Bloomology wrote, language as cause of human affection, please?

Affliction.

Affliction.

Is there possibly a connection with speech being disembodied?

Speech being disembodied.

I mean, I struggle to find a connection really.

I mean, speech is disembodied.

Is that referring to the fact words are kind of arbitrary sequences of sounds and don't have any resemblance to the things they're talking about?

I think if that's the question, that words are totally decoupled from their meaning in terms of the sound sequence has nothing to do with what the word means.

I think that feature of language arose because we need so many words.

And we need words for things which there's no possible way in which the sound can resemble the concept.

And so we just need some arbitrary string of sounds to represent some concept.

Again, I'm not, I feel I'm not answering the question here.

Yes, that reminds me of kind of Schrodinger's What is Life and about the quasi periodic crystal and about the way that there, there, there has to be some substrate that has enough reuse and modularity to allow recomposition.

But then that, to the extent that those compositions are meaningful, and this is also kind of explored in the Gerdel-Escher Bach, the subunits do not have that type of meaning.

Yeah, that's right.

Letters mean things that words can't, words must do and mean things that sentences can't, and like they, they partition into this hierarchical Bayesian niche.

And so then that creates these smaller artifacts which need to be taken in a Gestalt context.

Yeah, they have to be arbitrary in some way.

Yeah, and that defies sort of reductionism.

We don't, we don't take a magnifying glass and look deeper at the pixels on the paper.

But how about this human sense of self is secondhand and flawed.

So, we're having a first-hand experience of that secondhand flawed concept.

So, what is the first-hand experience if the secondhand is flawed?

The first-hand, the first-hand concept of self would be me thinking about myself directly.

I do this, I do that, and therefore I am that way or this way.

The second-hand concept is me not thinking about me as my own personal experience, but me trying to imagine what other people think of me.

And me trying to guess, does my bum look big in these trousers sort of thing?

What do I look like to other people?

That's the secondhand part of it.

It's not my, my idea of me.

It's my idea of what other people are thinking of me and why it's flawed.

One reason is that I'm imagining what other people think of me when other people are actually not thinking about me.

They're just thinking about themselves.

And I'm also doing it by these quick Bayesian patterns, pattern matching, which are sometimes inaccurate.

People's idea of themselves, as we know, can be very inaccurate.

So, that's why I say it's secondhand flawed.

So, that's kind of like the thinking through other, other minds, TTOM.

Why is that flawed when we could also have a first-hand experience that has false positive or false negative assertions?

Well, it's secondhand that it's in a mirror, the mirror of other people.

It's flawed in that when we learn these patterns in conversation, the self patterns, status patterns, we are only learning what we need to carry on a conversation.

If once we've learned something good enough to carry on a conversation with somebody, that's it.

It needn't learn any more.

But that may not be an accurate view of what they really think of us.

We're extrapolating from, in conversation, we kind of direct evidence of what the other person's thinking because they, if we say something wrong, they, or something that's based on a misconception, they will correct us.

But the shadow audience in private thought can never correct us.

So, we can get away with errors.

We can apply status patterns that are just wrong in private thought and nobody will ever correct us.

Whereas in public conversation, people will sometimes correct us.

Interesting, because that shadow audience can also be a sort of critical internal commentary.

So, it can make critical comments.

Oh yeah, absolutely.

It can drive that.

It can drive that.

I mean, it can lead to internal conscious experience of, you can, it's pre-conscious, it's pre-conscious pattern matching, but it can lead to conscious trying to make it better, trying to work out what other people will think of

you.

But the conscious stuff is always going so much slower than the pre-conscious pattern matching.

So, the pattern matching is going at two or three patterns, status patterns per second.

Conscious thought is going one new thought every five seconds.

Yeah.

I'm very interested in that.

Like, what kinds of exercises or experiments help us get a look at that fine-toothed gear or that kind of splicing of the moment that happens at the sub-second level?

How do you, how over your life and research have you explored that and how did it come to be the way that you think of it now?

Well, I don't think we really have a microscope to look at our own thinking at the sub-second level.

So, this is kind of theoretical speculation.

It's saying how language works.

I mean, I know that when I hear somebody speaking at three words per second, I'm doing three fast Bayesian pattern matches per second to understand what he's saying.

I'm doing more than that because I'm trying out several other patterns which don't make the cut, so to speak.

So, I know it happens, but I can't look at myself internally and say what's happening at one-fifth or second interval.

The only thing I can do towards that is practice mindfulness when I slow down some of this stuff and perhaps get some other insights into what's going on.

I mean, I think in mindfulness you do experience life without the urge to do things and life without the urge to change things.

And you can bring insights out of that which are valuable to you.

This is what happens.

But, no, there's no way of putting a microscope on your own thought.

How do you relate that to neuro-phenomenology and to cognitive science and what kinds of tools or what apparatus measurement or perturbation apparatuses do you think would further this line?

Well, I mean, in the lab we could do MRI experiments or to actually find out.

I mean, perhaps we could actually get an experimental probe of Bayesian pattern matching of words in the lab and see these things happening externally in a person in a scanner.

But that's a speculation. We're not very close to that yet.

I think. Well, we see what parts of the brain language is happening.

But to get the detailed patterns about how status patterns are happening, for instance, would be it's possible, but not happened yet.

Where do you see this, these research questions going or taking you over the coming years for yourself and for the field?

Like, what are we seeking to reduce our uncertainty about?

What would be the learning moments and what would be the utilities?

Well, firstly, the learning moments. What are we trying to reduce our uncertainty about?

I think we're still trying to get a decent model of language.

I think the world has been very confused by Chomsky and linguistics, which has introduced a load of red herrings.

I think the picture of cognitive linguistics needs to be reinforced.

And I think in research terms, as I've tried to do in these talks, you need to extend cognitive linguistics beyond syntax and semantics to do pragmatics.

I think we need to validate a good model of pragmatics by testing it against lots and lots of pragmatic phenomena and building those models computationally that work.

And that's what I'm going to be trying to do, build a working computational model of pragmatics.

So I think there's a lot we can do to improve our theoretical understanding of language.

Now, where was I going with that?

Oh, yeah. In terms of the practical impact, I think there is potentially huge practical impact because there is so much suffering and bad things happening in the world as a result of the shadow audiences and a result of status patterns.

And mindfulness does offer some relief from that.

So I think a better scientific understanding of mindfulness might in time lead to better ways of learning it and teaching it and more widespread mindfulness, more widespread appreciation of why the big bad things go wrong in the world.

And who knows, we might eventually be able to collaborate with each other enough to solve problems like climate change.

So there are huge practical implications if mindfulness is a way forward on these afflictions of the human race.

So I think that's a way to embrace mindfulness.

Yeah, there's a lot there in the the scale of coordination that difference communication and trust mechanisms enable.

enable. Several of these pieces are about sexual selection and about that sort of, about how even though the sexual environment is natural, it still has selective signatures that can be distinguished from the abiotic niche with selection operating on it.

And I guess what, where does that line take us understanding that although language helps us address these basic biotic challenges, it is also able to be harnessed by mimetic or even, yeah, transient peacock type dynamics or green beard dynamics, all these different more evolutionary sub games that exist. Like, is that our, our fated condition or would there be a way to...

Well, I, I mean, I don't think evolutionary timescales are of interest here. And those evolutionary timescales are at least thousands of years before the human species changes in any perceptible way. I think we've got much less time than that. I think we need to understand how the human race shoots itself in the foot repeatedly over the next 50 or 100 years and start getting an understanding of why we do that and how we can do it less out into common understanding, not esoteric scientific understanding, much more widely in the next 100 years, so that people

are less beset by these status issues and tribalism and groupthink and so on and so forth. In the general public, the whole 9 billion people on the planet, a large proportion of them should be somehow see mindfulness as less esoteric, less mystical and the practical common sense thing to live better and more cooperatively. I don't know how that's going to happen or whether it's going to happen, but it is a possibility.

In the chat, what does modelling pragmatics with others look like? What does which pragmatics? What does modelling pragmatics with others look like? Like possibly what is that in practice, what kinds of conversational games or exercises or trainings or experiences could help bring a fluency to those who are exploring what it means to model pragmatics of language together? How do people bring this into their daily conversations? Is it...

I don't think they do. I think it's a matter of understanding how we do daily conversation, not trying to manipulate that in any way. In other words, what we do in conversation...

Well, let me step back a bit. What we do in conversation is a product of this learning conditioning process where we learn all these word patterns, we learn status patterns. I think one very important change mechanism is to... is really with young children when they're learning. And maybe there are ways in which adults can structure the environment, just bring children up better so that they don't learn the harmful status patterns. And I don't think that is in the structure of conversations. It's in understanding where self-esteem is.

where self-esteem comes from, understanding how you give a child space in which to learn about things and not learn the harmful self-esteem patterns. So I think that is a tremendous potential for progress. If we understand better how to bring up our children and do so, so they're more sane than we were.

I'm also sometimes struck by that with bringing cognitive science into day to day, but there isn't anything else to do or bring in. Any activity is going to fall into this category.

It's more of a lens that helps us understand any kinds of speech acts.

I think that's right. It's a lens that helps us understand rather than manipulate or change.

Language is about status is still so interesting to me as well. Because seemingly, apparently there are expressions that don't relate to status.

Are those just accessory expressions that are kind of tributary, kind of corollary expressions that don't really arise from the main thrust?

Or do we translate even the status personification,

personification,

objectification, ranking,

judging,

on to even like,

well, this is my favorite coin on the table.

And basically applies status games to even objects.

Anything we can talk about

can have possible status reverberations in some people who hear it.

You know, you can't.

No, so I mean, the basic idea is that people have usually thought of language as some kind of neutral medium, harmless medium, and all benefits medium, and all benefits medium, which we use to talk about reality.

And it's neutral and it doesn't bias our view of reality.

What I'm saying is that really it's so soaked in status as in this slide.

I'll come back to this slide here.

Language is so soaked in status that more or less any sentence you say is going to bring in status issues in some listeners.

And it's going to reverberate with them.

And more or less anything you say can, even when you're talking about some abstract scientific topic, if you're favoring a scientific theory or hypothesis, which somebody else has worked on a different hypothesis, that immediately reverberates with their self-esteem.

They don't like it when different ideas challenge their own.

So,

speech acts are political.

Actually, this slide that you have right here, this relates to sort of the man as political animal, and about the distinction between zoo and bios, and like zoology and biology, and about different...

There's just a lot of interesting etymology and theory there that Agamben and others have explored about.

The zoo, like zoology, is this right column with the animal layer.

And then the layer of what is added beyond that, that makes, you know, human the political animal.

Yeah, I mean, I could have used the word politics in many places in this talk.

Yeah, it comes in.

Politics is just status written large, really.

Hmm.

Yeah, what are your next moves?

Or what...

Let's check in.

What is on your research portfolio?

We have the wave, the flies, consciousness, like, how are you seeing these different threads merging and splitting right now?

Well, I think one important strand is to merge the wave consciousness idea with this model of language.

And I think a lot may come out of that.

I would like to sort of integrate this more with the FEP portfolio of ideas.

And if anybody is interested in doing an FEP, because this is...

I'm saying language is Bayesian inference.

And I'm saying that the wave is a Bayesian optimal model of our physical surroundings.

And so they're all very FEP compatible.

And I would like to do FEP implementation of it.

If somebody who's more expert than I am in FEP techniques would care to collaborate on it.

So I think those are important lines.

But integrating language and consciousness, I think, is very important.

Because a part I haven't talked about here is how we get conscious of language meanings.

And that all has to be done.

I think just mechanizing the model of pragmatics, producing computer implementation that does pragmatics, and demonstrating the steps of building up the common ground and so on.

That's an important topic.

So there's plenty going on.

Where would you say pheromonal symbolic communication would come into play in reference to the insect neuroanatomy or not?

Now I'm currently looking at insect brain anatomy in a very different way.

Because obviously they have language or anything like it.

They probably do have consciousness.

But what's happening on the insect brain with me at the moment is I'm just looking at the data on the fruit fly connectome and finding patterns, which I don't think have been seen before, groups of neurons that act together.

And I'm just looking at the data on the other, where I'm called partner neurons.

And this is just saying to me that there are things about brains in general, insect brains or vertebrate brains, which are just much more complicated than we think.

I'm beginning to think the old Macaulic-Pitts model of a neuron, where neurons are kind of simple adding machine.

Can't be the whole story, only a tiny part of the story.

And I think insect brains may help us because they're such, because they are smaller and more examinable.

They may help us explore that brains are really doing.

I think the neuron, if you think a neuron is just a Macaulic-Pitts neuron with Hebbian learning, that seems to me such a vastly oversimplified picture of what a neuron is, but it can't be right.

And that's what I'm trying to look at with the insect brains.

But that's a different topic, really.

Yeah.

Oh, well, yeah.

A lot of cool things with intracellular computation, dendritic trees.

And when you look at the morphology of some of these neurons, amazing in the insect brain.

But they do also have different songs.

Different?

Different?

Sometimes songs, like different stridulation.

Like language.

It's, what would you say that it has to be oral?

Or could it be sequential in it?

Or does it have to even be sequential, even something like Braille?

I mean, is the key aspect that is...

I think we should...

Yeah.

I think we should just restrict the word language to human language.

And we should use a different word, like signalling or something, for what animals do.

And insects certainly do signalling, but they've never had sexual selection for language ability.

Never had to show off their intelligence to other insects, I don't think.

Yeah.

What about sexual signalling with song and dance and nest construction?

Oh yeah.

There's a huge amount of that all over nature.

People, animals learn courting rituals and courting songs and so on and that.

But none of those, I think, have got to the state of human language of showing off.

Animals have to show off that they know the code.

They know how to dance or they know how to sing a certain pattern of songs.

But they don't have to show off that they're more intelligent.

They can compose these things using intelligence.

So I think that's the specific, the human thing which led to our oversized brains.

So what ways would you say people could get involved or contribute or just how do you take this from here?

Well, I'm hoping to do is get the pragmatics ideas more widely appreciated.

And if anybody wants to pick those up and do an FEP implementation of them or work with me on a general implementation, that would be great.

But I just want to get those ideas out there.

Because what I feel is that if I present the unified theory of language, that's too much to take in one bite.

And people will say, oh, well, there's too much there.

I haven't got time to read it.

Whereas if I can concentrate just on pragmatics, people might say, let's have a look at this.

Just want to get the ideas out there.

Yeah, I'll be really interested to see that.

And Elliot Murphy's work, Similarities and Differences, and Michael Walker and Hector Manrique and their discussions of kind of the cognitive niche and the communicative niche that led to theory of mind type developments.

But you brought up many interesting areas.

And I mean, there's a lot to.

How can we wordlessly or wordfully integrate?

Let's just say that after this stream, we were going to reflect.

Do we always return to the same mindful attractor with generic cues to release or evade linguistic naming?

Or is there a way to set an orientation for, let's just say, material heard during a lecture that both points towards the content that was learned,

but also doesn't stay within a linguistic interior monologue rehearsal setting?

That's huge questions there.

I come back to mindfulness.

I think a big part of mindfulness is being mindful of the fact that you're not mindful, being aware of these patterns dragging on you.

And that, in a strange way, leads to more freedom from them.

But it's hugely difficult to summarize, especially in words.

Do you feel like any of this has to do with your name being Warden?

Well, it's an old Norman name.

It means guardian.

I don't think.

Very well.

Very well.

Thank you, Robert.

Looking forward to how this proceeds in exploring the insects and hearing more about consciousness at the symposium.

Great.

Many thanks, Dan.

See ya.

Bye.

See ya.

Bye.

See ya.

See ya.